Derivada de 2 variables. Ej. continuación (1)

(1) $Z(x,y) = x^3 . \operatorname{sen} x^2 y^3$

$\frac{\partial z}{\partial x} = 3x^2 \operatorname{sen} x^2y^3 + x^3 y^3 2x \cos x^2 y^3$

$\frac{\partial z}{\partial x} = 3x^2 \operatorname{sen}(x^2y^3) + 2x^4y^3 \cos(x^2y^3)$

$\frac{\partial z}{\partial x \partial y} = 3x^2 x^2 3y^2 \cos(x^2y^3) + 2x^4 3y^2 \cos(x^2y^3) + 2x^4y^3x^2 3y^2 \left[-\operatorname{sen}(x^2y^3)\right]$

$6x^6y^5 \operatorname{sen}(x^2y^3)$

$\frac{\partial z}{\partial x \partial y} = 9x^4y^2\cos(x^2y^3) + 6x^4y^2\cos(x^2y^3) - 6x^6y^5\operatorname{sen}(x^2y^3)$

$\frac{\partial z}{\partial y} = x^3x^2 3y^2\cos(x^2y^3) = 3x^5y^2\cos(x^2y^3)$

$3y^2 5x^4 \cos(x^2y^3) + 3x^5y^2y^3 2x\left[-\operatorname{sen}(x^2y^3)\right]$

$\frac{\partial z}{\partial y \partial x} = 3y^2 5x^4 \cos(x^2y^3) + 3x^5y^2x^2.3y^2\left[-\operatorname{sen}(x^2y^3)\right]$

$\frac{\partial z^2}{\partial y^2} = 3x^5 2y\cos(x^2y^3)$

Tabla

$y = x^n$

$y' = n x^{n-1}$

$y = u.v$

$y' = u'v + uv'$

$y = \operatorname{sen} u$

$y' = u' \cos u$

$y = \cos u$

$y' = u'(-\operatorname{sen} u)$

$y = e^u$

$y' = u'. e^u$

$y = u.v.w$

$y' = u'vw + v'uw + w'vu$

(2) $Z(x,y) = \operatorname{sen}(x^2y^{-3}) . e^{x^3y^4}$

$\frac{\partial z}{\partial x} = y^{-3} 2x \cos(x^2y^{-3}) . e^{x^3y^4} + \operatorname{sen}(x^2y^{-3})\, y^4 3x^2 e^{x^3y^4}$

$\frac{\partial z}{\partial y} = -3y^{-4}x^2\cos(x^2y^{-3}) . e^{x^3y^4} + \operatorname{sen}(x^2y^{-3}) . x^3 4y^3 e^{x^3y^4}$

(3) $z(x,y) = \operatorname{sen} y . \cos(2x^2y)$

$$\frac{\partial z}{\partial x} = \operatorname{sen} y \cdot y\, 4x\left[-\operatorname{sen}(2x^2y)\right] = \underbrace{-4xy}_{1}\ \underbrace{\operatorname{sen} y}_{2} \cdot \underbrace{\operatorname{sen}(2x^2y)}_{3}\ 2$$

$$\frac{\partial z}{\partial x\,\partial y} = -4x \operatorname{sen} y \operatorname{sen}(2x^2y) + \cos y\,(-4xy) \cdot \operatorname{sen}(2x^2y) + \cos(2x^2y)\, 2x^2(-4xy) \cdot \operatorname{sen} y$$

$$\frac{\partial z}{\partial x\partial y} = -4x \operatorname{sen} y \operatorname{sen}(2x^2y) - 4xy \cos y \operatorname{sen}(2x^2y) - 8x^3y \operatorname{sen} y \cos(2x^2y)$$

$$\frac{\partial z}{\partial y} = \cos y \cos(2x^2y) + \operatorname{sen} y \cdot 2x^2\left[-\operatorname{sen}(2x^2y)\right]$$

$$\frac{\partial z}{\partial y} = \underbrace{\cos y \cos(2x^2y)}_{1} - \underbrace{2x^2 \operatorname{sen} y}_{1}\ \underbrace{\operatorname{sen}(2x^2y)}_{2}$$

$$\frac{\partial z}{\partial y\,\partial x} = \cos y\ 4xy\left[-\operatorname{sen}(2x^2y)\right] - \left[4x \operatorname{sen} y \operatorname{sen}(2x^2y) + 2x^2 \cdot \operatorname{sen} y \cdot 4xy \cos(2x^2y)\right]$$

④ $$z(x,y) = \frac{3x^{-2}y^{-3}}{\operatorname{sen}(x^2y^5)}$$

$$\frac{\partial z}{\partial x} = \frac{-6x^{-3}y^{-3} \cdot \operatorname{sen}(x^2y^5) - 3x^{-2}y^{-3} \cdot 2xy^5 \cos(x^2y^5)}{\left[\operatorname{sen}(x^2y^5)\right]^2}$$